Introduction to Data Analytics

Module 1 : Introduction to Data Analytics

1. Introduction to Data Science
2. Big Data and Data Science
3. Introduction to Big Data Platform
4. Challenges of Conventional Systems
5. Intelligent data analysis
6. Nature of Data
7. Analytic Processes and Tools
8. Analysis vs Reporting
9. Modern Data Analytic Tools

Module 2 : Data Representation and pre-processing

1. Multi-Dimensional data
2. Basic tools (plots, graphs and summary statistics) of EDA
3. Needs Preprocessing the Data
4. Data Cleaning
5. Data Integration and Transformation
6. Data Reduction
7. Discretization and Concept Hierarchy Generation.
8. Data summarization
9. Data Normalization.

Module 3 : Machine Learning Concepts

1. Statistical Inference - Populations and samples - Statistical modeling
2. probability distributions
3. fitting a model
4. Feature Generation (role of domain expertise) and Feature Selection algorithms: Filters; Wrappers

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Module 4 : Supervised and Unsupervised Learning

1. Decision Trees
2. Random Forests,
3. Machine Learning Algorithms - Linear Regression
4. - k-Nearest Neighbors (k-NN)
5. k-means
6. Naive Bayes
7. SVM
8. Clustering
9. Expectation Maximization
10. Dimensionality Reduction
11. Feature Selection
12. PCA
13. factor analysis

Module5 : Reinforcement Learning

1. Reinforcement Learning: Value iteration
2. policy iteration
3. TD learning
4. Q learning
5. actor-critic